N+ Network Security

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Assignment-13

Prepare By
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1. **Simulate a DNS resolution failure on your local machine by editing the hosts file to point www.spotify.com to a wrong IP address, then try to ping the domain and document the error message you receive.**

Aim:

To simulate a DNS resolution failure by editing the Windows hosts file and pointing www.spotify.com to an incorrect IP address.

Procedure:

1. Opened Notepad as Administrator.
2. Opened the hosts file from:

C:\Windows\System32\drivers\etc\hosts

3. Added the following entry:

192.0.2.1 www.spotify.com

4. Saved the hosts file.
5. Opened CMD and executed:

ipconfig /flushdns

6. Pinged the domain:

ping www.spotify.com

Expected Error Output:

Pinging www.spotify.com [192.0.2.1]

Request timed out.

Result:

Successfully simulated a DNS resolution problem by mapping www.spotify.com to an incorrect IP address in the hosts file. The ping command did not reach the actual Spotify server.

2. Configure a DHCP server in a network simulator (such as Cisco Packet Tracer or GNS3) and intentionally set an incorrect default gateway; connect a client device and verify the assigned gateway, then describe how you would identify and fix this issue.

Aim:

To configure a DHCP server with an incorrect default gateway and identify and fix the issue.

Procedure:

1. Opened Cisco Packet Tracer and created a network with one router, one switch, one DHCP server, and one PC.
2. Configured the DHCP server with the following settings:

Setting	Value
IP Address	192.168.1.2
Subnet Mask	255.255.255.0
Incorrect Gateway	192.168.1.254
DHCP Start IP	192.168.1.10

3. Enabled the DHCP service.
4. Connected the PC to the network and selected DHCP in IP Configuration.
5. Checked the assigned IP address and default gateway.
6. Used the following command to verify the configuration:

ipconfig /all

7. Identified that the assigned default gateway was incorrect.
8. Corrected the DHCP pool gateway to 192.168.1.1.
9. Renewed the IP address using:

ipconfig /release

ipconfig /renew

Result:

Successfully identified and corrected the incorrect default gateway in the DHCP configuration. The client received the correct gateway address from the DHCP server.

3. Given a scenario where devices on your simulated network are unable to access websites by name but can by IP address, use the nslookup or dig tool to diagnose the DNS problem and write down the troubleshooting steps and solution.

Aim:

To diagnose and troubleshoot a DNS problem where devices can access websites by IP address but not by domain name.

Troubleshooting Steps:

1. Checked internet connectivity by pinging a public IP address:

   ping 8.8.8.8

2. Tested DNS resolution using nslookup:

   nslookup www.google.com

3. Found that DNS resolution was failing because of an incorrect DNS server address.
4. Checked the DNS server configuration using:

   ipconfig /all

5. Corrected the DNS server address to 8.8.8.8.
6. Cleared the DNS cache using:

   ipconfig /flushdns

7. Tested DNS resolution again:

   nslookup www.google.com

8. Verified website access using the domain name.

Solution:

Configured the correct DNS server address (8.8.8.8) and cleared the DNS cache.

Result:

Successfully diagnosed and fixed the DNS resolution problem. Devices could access websites using domain names.

4. A device in your simulated network is not receiving an IP address from the DHCP server and shows an APIPA address (169.x.x.x). List the possible causes for this and document the step-by-step process you would use to resolve it.Hint: Check for network connectivity, DHCP scope exhaustion, and server availability.

Aim:

To identify and troubleshoot a device that receives an APIPA address (169.x.x.x) instead of an IP address from the DHCP server.

Possible Causes:

1. DHCP server is turned off or unavailable.
2. DHCP scope has no free IP addresses.
3. Network cable or switch connection is faulty.
4. Incorrect DHCP configuration.
5. DHCP service is disabled.

Troubleshooting Steps:

1. **Check Network Connectivity:** Verify the network cable and switch connection.
2. **Check IP Configuration:** Run the command:

ipconfig /all

3. **Check DHCP Server:** Ensure that the DHCP server is running and connected to the network.
4. **Check DHCP Scope:** Verify that the DHCP pool has available IP addresses.

5. **Check DHCP Configuration:** Verify the IP range, subnet mask, and default gateway.
6. **Renew IP Address:** Execute:

ipconfig /release

ipconfig /renew

7. **Test Connectivity:** Ping the DHCP server to check network connectivity.

Solution:

Restart the DHCP service, correct the DHCP configuration, ensure free IP addresses are available, and check network connectivity.

Result:

Successfully identified and resolved the DHCP issue so that the device could receive a valid IP address from the DHCP server.

5. Create a table documenting at least three common DNS issues and three common DHCP issues you encountered or researched, including symptoms, possible causes, and recommended solutions for each.

Common DNS and DHCP Issues

1. Common DNS Issues

DNS Issue	Symptoms	Possible Causes	Recommended Solutions
DNS Resolution Failure	Website does not open by name.	Incorrect DNS server IP.	Configure the correct DNS server.
Incorrect DNS Record	Domain points to the wrong IP address.	Wrong A Record configuration.	Correct the DNS A Record.
DNS Server Unavailable	Websites cannot be accessed by name.	DNS service is stopped.	Restart the DNS service.

2. Common DHCP Issues

DHCP Issue	Symptoms	Possible Causes	Recommended Solutions
APIPA Address	Device receives 169.x.x.x IP address.	DHCP server is unavailable.	Check and restart DHCP service.
IP Address Conflict	Two devices have the same IP address.	Duplicate IP configuration.	Assign unique IP addresses.
DHCP Scope Exhaustion	New devices cannot get IP addresses.	DHCP pool has no free IPs.	Increase the DHCP IP address range.

Result:

Successfully identified common DNS and DHCP issues, their possible causes, and recommended solutions for network troubleshooting.